Fourier Series and Fourier Transform

Participated in & did the coursework [Y/N]?

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1 Introduction

Analog filters serve as foundational tools in signal processing. They are mathematically constructed to isolate specific frequency components by passing desired signals while attenuating out-of-band noise. When a system requires the extraction of a strict frequency range—a common requirement when isolating faint target signals from overwhelming ambient noise—a bandpass configuration is typically employed. Filter performance relies heavily on the behavior within the passband, where signal loss should ideally be minimal, and the stopbands, which must provide sufficient suppression of unwanted frequencies. This laboratory exercise focuses on the design, synthesis, and comparative evaluation of four classical analog bandpass approximations: the Butterworth, Chebyshev Type I, Chebyshev Type II, and Elliptic models. The primary goal is to determine how well these architectures can meet a stringent set of specifications. These include passband edges defined at 2000 and 4000 rad/sec with an allowable ripple of 1 dB, alongside stopband edges at 1500 and 4500 rad/sec that demand at least 60 dB of attenuation. To establish a comparative baseline, the physical filter models are evaluated against a mathematically ideal, though physically unrealizable, bandpass response.

Executing this design relies on a systematic computational approach implemented in MATLAB. The initial theoretical step requires calculating the minimum filter order and corresponding natural frequencies necessary to satisfy the required attenuation limits. This is accomplished using specific order-estimation functions (`buttord`, `cheb1ord`, `cheb2ord`, and `ellipord`). Once the optimal order is established, the transfer functions are synthesized in the continuous s-domain by generating their respective numerator and denominator coefficients. Evaluating the frequency response across the target spectrum then involves computing the complex response via the `freqs` function. From there, magnitude and phase data are extracted and converted into decibels and degrees for standard analysis. As a final analytical step, an ideal bandpass filter is constructed using symbolic variables and the Heaviside step function. This theoretical model acts as a visual and mathematical reference point to better highlight the inherent limitations of the synthesized approximations.

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2 Experimental Design

The primary objective of this experiment is to design, simulate, and compare four major types of analog filter approximations: Butterworth, Chebyshev Type 1, Chebyshev Type 2, and Elliptic filters. The goal is to determine how well each mathematical model adheres to a specific set of bandpass constraints and how they deviate from an ideal filter response. This methodology utilizes a computational approach to determine the minimum required filter order, synthesize the transfer functions in the s-domain, and transform those functions into the frequency domain for visual analysis of magnitude and phase.

2.1 Technical Specifications and Constraints

The design process is governed by a set of rigid parameters that define the behavior of the bandpass filter. The passband, which is the range of frequencies allowed to pass through the filter, is set between a lower edge of 2000 rad/sec and an upper edge of 4000 rad/sec. Within this region, a maximum ripple of 1 dB is permitted. The stopbands, where the signal must be significantly blocked, are defined as frequencies below 1500 rad/sec and above 4500 rad/sec. In these regions, the filter must provide at least 60 dB of attenuation. These requirements serve as the foundation for calculating the mathematical complexity of each filter.

Design Parameter	Symbol	Target Value
Lower Passband Frequency	W_{p1}	2000 rad/sec
Upper Passband Frequency	W_{p2}	4000 rad/sec
Lower Stopband Frequency	W_{s1}	1500 rad/sec
Upper Stopband Frequency	W_{s2}	4500 rad/sec
Maximum Passband Loss	R_p	1 dB
Minimum Stopband Attenuation	R_s	60 dB

Table 1
Required Filter Performance Specifications

2.2 Mathematical Modeling and Order Calculation

In analog filter design, the system is represented by a transfer function $H(s)$ in the Laplace domain, which is a ratio of two polynomials. The complexity of the filter is defined by its "order"

(N); a higher order results in a steeper transition between the passband and stopband but increases the mathematical and physical complexity of the system.

The first step in the procedure involves using specialized algorithms, such as `buttord` and `ellipord`, to calculate the lowest integer order N that satisfies the 1 dB ripple and 60 dB attenuation requirements. These functions analyze the "steepness" required to drop from the passband to the stopband. Once N is found, the specific coefficients for the numerator and denominator of the transfer function are generated using standard approximation formulas for each filter type.

2.3 Frequency Response Transformation

To evaluate how these filters perform across a real-world spectrum, the s-domain transfer function must be evaluated as a function of angular frequency ($j\omega$). The experiment defines a sample range from 500 to 6000 rad/sec. By applying the `freqs` function, the theoretical Laplace expression is converted into a complex frequency response. From this result, the magnitude is derived by taking the absolute value, allowing us to see how much of the signal is kept or blocked. The phase is derived by calculating the angle of the response in radians, which illustrates how the filter shifts the timing of the input signal.

Finally, an ideal filter is mathematically constructed to serve as a benchmark. This ideal model is defined to have a magnitude of exactly 1 within the 2000–4000 rad/sec range and 0 everywhere else, with a perfectly flat phase. Comparing the four analog designs to this ideal model reveals the inherent trade-offs of each approximation, such as the Butterworth's smoothness versus the Elliptic's rapid roll-off.

2.4 Procedural Logic (Pseudocode)

The following logic outlines the step-by-step execution used to produce the experimental data and corresponding visualizations:

```
1.  START
2.  INITIALIZE design constants:
        Wp = [2000, 4000], Ws = [1500, 4500]
        Rp = 1, Rs = 60
        Frequency_Range (w) = 500 to 6000 step 10
3.  FOR EACH Approximation (Butterworth, Cheby1, Cheby2, Elliptic)
        a. Determine Order (N) and Cutoff (Wn) using 'ord' tools
        b. Generate transfer function coefficients (b, a)
        c. Transform (b, a) to frequency response (h) using 'freqs'
        d. Magnitude_Data = abs(h)
        e. Phase_Data = angle(h)
        f. Add Magnitude_Data to Figure 1
        g. Add Phase_Data to Figure 2
    END FOR
4.  CREATE Ideal Filter:
        IF (w >= 2000 AND w <= 4000) THEN Magnitude = 1
        ELSE Magnitude = 0
        Phase = 0 for all w
5.  OVERLAY Ideal Filter data on Figures 1 and 2
6.  LABEL axes and APPLY legends
7.  END
```

3 Analysis of Results

The results obtained from this exercise provide a comprehensive understanding of frequency-domain representations and system filtering behavior. To ensure computational reliability and mathematical validity of the exercise, independent simulations are consolidated and verified.

3.1 Fourier Series Decomposition and Magnitude Spectrum

A periodic signal can be broken down into individual frequency components having the magnitude spectrum showing how the signal's energy is distributed across the mentioned frequencies. The frequency-domain representation of the given periodic sawtooth signal $x_p(t) = -2t$ with a period of $T = 0.75$ quantifies energy distribution. Through calculating the exponential Fourier series coefficients (C_n) the analytical composition of the signal is found. The resulting magnitude spectrum which was plotted for the harmonic index $-10 \leq n \leq 10$ reveals a symmetric decay centered around the fundamental DC component. This peak shows that the lowest frequencies hold the majority of the waveform's power while the higher harmonics have less. Just to maintain computational stability throughout numerical evaluation of this an infinitesimal variable (`eps`) was required to prevent division-by-zero singularities at the exact zero-frequency index ($n = 0$).

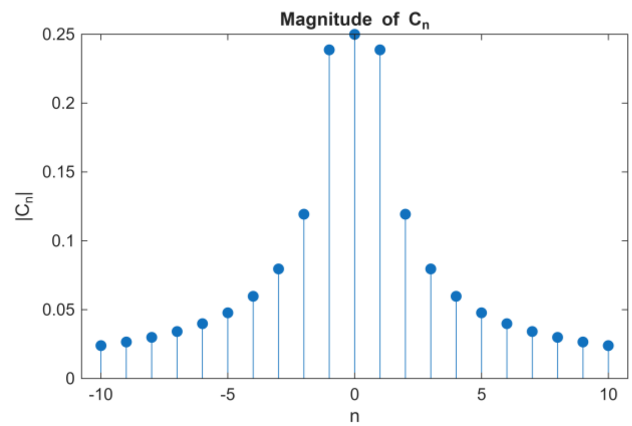

Figure 1. Magnitude spectrum of the exponential Fourier series coefficients for $-10 \leq n \leq 10$, illustrating peak energy concentration at the fundamental frequency.

3.2 Signal Reconstruction and the Gibbs Phenomenon

Reconstructing a signal that contains amplitude changes using a limited number of frequencies results in a behavior known as the Gibbs phenomenon. Generating the analytical expression for the truncated signal $x_{pN}(t)$ by utilizing the computed finite series of 21 harmonics produces algebraic complexity and the time-domain also needs numerical substitution over time vector $\Delta t = 0.001$. Overlaying this reconstructed sequence against the ideal original signal $x_p(t)$ shows the mathematical limitations of finite harmonic summation while the approximation accurately follows the linear downward slope of the sawtooth wave. This exhibits significant structural deviations at the interval boundaries.

3.3 Transfer Function and Cut-off Linearity

A system's transfer function dictates its frequency filtering behavior and the 3dB cut-off frequency in the exercise defines the exact boundary where the transmitted signal loses half of its power. For aperiodic continuous-time systems the exponential impulse response $h(t) = \frac{a}{2}e^{-a|t|}$ represents an ideal low-pass filter. Deriving the transfer function $H(\omega)$ using continuous Fourier transform establishes the baseline frequency-domain behavior. Analytically solving for the 3dB cut-off frequency yields multiple algebraic roots due to the quadratic nature of the frequency variable and getting the magnitude establishes the exact mathematical relationship $\omega_c = a\sqrt{\sqrt{2}-1}$. Evaluating

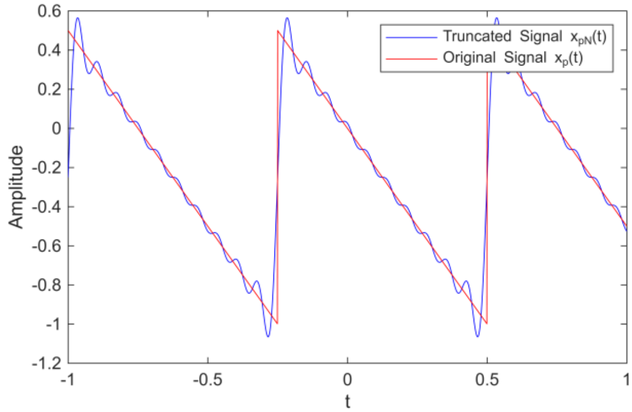

Figure 2. Time-domain comparison of the original sawtooth signal and its truncated Fourier reconstruction, highlighting oscillatory ringing at waveform discontinuities.

and plotting that magnitude expression through $0 \leq a \leq 100$ generates a strictly positive linear plot. This graphical result mathematically proves a direct proportional dependency with a scaling constant of approximately 0.6436 where the operational frequency boundary of the filter is defined by its exponential scaling factor a.

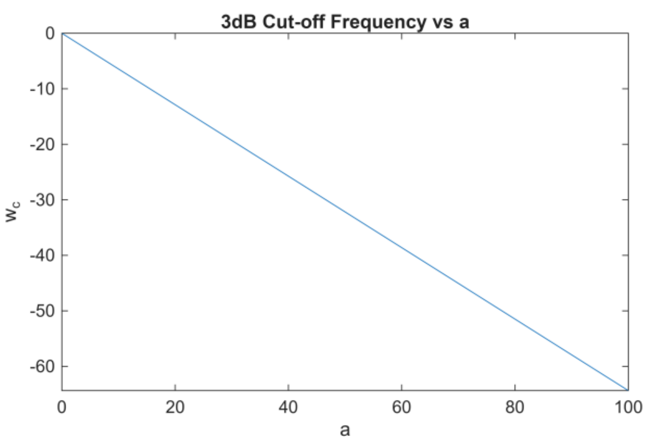

Figure 3. Linear proportional relationship between the exponential system parameter a and the resulting positive 3dB cut-off frequency magnitude ω_c.

3.4 Parametric Bandwidth Expansion

Increasing a low-pass filter's cut-off frequency naturally widens its passband that allowing a broader range of frequencies to pass through the system unattenuated. The practical implications of the previously established linear relationship are validated by evaluating the magnitude response curves $|H(\omega)|$ at discrete parametric intervals of $a = 10$, 50, 80, and 100. As the defining parameter a scales upward the attenuation slopes relaxes causing the bell-shaped passband to widen uniformly outward across the frequency axis. Consequently, higher cut-off frequencies natively expand the system's absolute bandwidth permitting a broader continuous spectrum of input frequencies to propagate through the system unattenuated [1].

4 Discussion and Conclusion

Based on the computational synthesis and spectral plotting, the results highlight the classical trade-offs that complicate

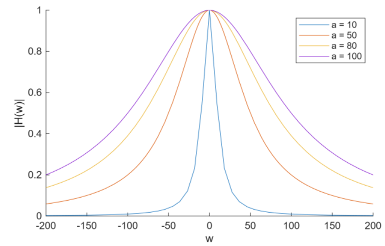

Figure 4. Superimposed magnitude responses demonstrating continuous bandwidth expansion as the exponential parameter a increases from 10 to 100.

analog filter design. The comparative analysis of the four filter architectures against the predefined specifications suggests that no single approximation offers a flawless solution. Instead, each model forces a compromise between transition steepness, passband flatness, and phase linearity.

Examining the magnitude response reveals distinct behavioral priorities among the designs. The Butterworth filter maintained a maximally flat passband without introducing ripple artifacts. This smoothness, however, comes at a cost: the gradual roll-off dictates a considerably higher filter order to reach the required 60 dB of stopband attenuation, which may pose challenges in physical implementation due to component sensitivity. The Chebyshev approaches present an alternative strategy. They achieve a steeper transition by permitting either a 1 dB ripple in the passband (Type I) or equiripple variations in the stopband (Type II). Taking this mathematical compromise to the extreme, the Elliptic filter emerges as the most order-efficient architecture. By distributing ripples across both the passband and the stopband, it yields the sharpest magnitude transition of the four models, though this aggressive filtering introduces complications elsewhere.

When correlating these magnitude profiles with the corresponding phase analysis, the limitations of physical approximations become even more apparent. While the symbolic reference model maintains a theoretical zero-phase shift, every realizable filter introduced some degree of non-linear phase distortion. The Butterworth filter managed to sustain the most linear phase response within its passband. This characteristic is generally critical for minimizing temporal signal dispersion, particularly when precise wave shape preservation is required. On the other hand, models designed for sharp magnitude cutoffs, most notably the Elliptic and Chebyshev variations, exhibited severe phase non-linearities. These distortions appear most pronounced near the designated cutoff frequencies of 2000 and 4000 rad/sec, suggesting that applications requiring high phase fidelity might need to avoid these steeper filters entirely or implement additional phase equalization stages.

Ultimately, the experiment demonstrates that practical analog filter design is largely an exercise in managing competing parameters. Selecting a specific approximation cannot be reduced to a single metric of performance. Rather, it depends heavily on the primary demands of the target system. An engineer must weigh whether the application fundamentally requires absolute magnitude flatness, highly efficient transition steepness, or the strict phase linearity necessary to preserve signal timing and integrity.

5 Authors Contribution

On what contributions to specify, see the terms at www.elsevier.com.

1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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[1] S. S. Haykin and B. Van Veen, *Signals and systems*. New York: Wiley, 1999.